

Mini Paper — 1.4 Data Types, Data Structures & Boolean Algebra — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Q1. [1+2 = 3] (a) $150 = 128 + 16 + 4 + 2 \rightarrow$ **1001 0110** (1) (b) Split into nibbles: **1001** = 9, **0110** = 6 $\rightarrow$ **0x96** (1 for nibble split, 1 for correct answer = 2) Check: $9 \times 16 + 6 = 144 + 6 = 150 \checkmark$ Examiner tip: always group binary into nibbles from the RIGHT before converting to hex.

Q2. [2] $2F \rightarrow$ first digit 2 = $(2 \times 16) = 32$ (1); second digit F = 15 $\rightarrow 32 + 15 = 47$ (1) Examiner tip: multiply the left hex digit by 16, then add the right digit's value (A–F = 10–15).

Q3. [4] (a) $45 = 0010\ 1101$; $58 = 0011\ 1010$ (1) (b) -58 : invert **0011 1010** $\rightarrow$ **1100 0101**, add 1 $\rightarrow$ **1100 0110** (1) (c) Add **0010 1101** + **1100 0110**:

```
  0 0 1 0 1 1 0 1   (45)
+ 1 1 0 0 0 1 1 0   (-58)
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  1 1 1 1 0 0 1 1
```

No carry out of the MSB. (1) Result **1111 0011** is negative (MSB = 1); magnitude = invert+1 = **0000 1101** = 13, so answer = **-13** (1). ($45 - 58 = -13 \checkmark$) Examiner tip: subtract by adding the two's complement; remember to discard any carry out of the MSB.

Q4. [3] (a) Mantissa must start **0.1...**; currently **0.00110**, so shift the point **right by 2** places $\rightarrow$ **0.110** (1). Moving the point right by 2 reduces the exponent by 2: $3 - 2 = 1$, giving **0.110 $\times 2^1$** (1). (b) Normalising removes wasted leading zeros so the mantissa holds the **maximum significant figures / greatest precision** for the available bits (1). Examiner tip: moving the point right **DECREASES** the exponent; the overall value must stay the same.

Q5. [3] (a) Arithmetic shift right by 1 on **1111 0100**: shift right and copy the sign bit (1) in from the left $\rightarrow$ **1111 1010** (1). (b) **1111 1010** = $-(\text{invert}+1) = -0000\ 0110 = -6$ (1). An arithmetic shift **preserves the sign bit**, so a negative two's complement number stays negative and is correctly divided by 2; a logical shift would insert a 0, corrupting the sign and giving a wrong (positive) result (1). Examiner tip: $-12 \div 2 = -6$ — state both the bit pattern and the denary check.

Q6. [4] - De Morgan's on the first term: $\neg(A + \bar{B}) = \neg A \cdot \neg(\bar{B}) = \bar{A} \cdot B$ (double negation of $\bar{B} \rightarrow B$) (1 for De Morgan's, 1 for double negation) - Expression becomes $(\bar{A} \cdot B) + (A \cdot B)$ (1) - Factor out B (distributive): $B \cdot (\bar{A} + A) = B \cdot 1 = B$ (1) Final answer: **B** Examiner tip: with De Morgan's, change the operator AND negate each term, then look for a common factor to absorb.

Q7. [3]

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

(1 mark per correct pair of rows = up to 2 for the column.) F is 1 only when the inputs **differ**, so it is equivalent to an **XOR** gate (1). Examiner tip: $A \cdot \bar{B} + \bar{A} \cdot B$ is the standard sum-of-products definition of XOR.

Q8. [4] (a) Tracing (size 5, MOD 5; rear points to next free slot):

Operation	Array (idx 0–4)	front	rear
enqueue P	[P, , , ,]	0	1
enqueue Q	[P, Q, , ,]	0	2
enqueue R	[P, Q, R, ,]	0	3
dequeue	[, Q, R, ,]	1	3
enqueue S	[, Q, R, S,]	1	4
enqueue T	[, Q, R, S, T]	1	0

Final array contains **Q, R, S, T** (index 0 free), **front = 1, rear = 0** (1 for correct contents, 1 for front, 1 for rear = 3). (b) A circular queue **reuses the space freed by dequeuing** (rear wraps round with MOD), so it does not waste slots at the front like a linear queue (1). Examiner tip: show the MOD wraparound — rear goes 4 → 0 once it passes the end of the array.

Q9. [4] (a) BST built by inserting 50, 30, 70, 20, 40, 60:

